

Authors' Responses

Journal Title: IEEE Robotics and Automation Letters

Reference Number of the Previous Submission: 25-2189

Title of Paper: Don't Let Your Robot be Harmful: Responsible Robotic Manipulation via Safety-as-Policy

Authors: Anonymous Authors

We would like to extend our heartfelt gratitude to the Editor-in-Chief and the Associate Editor for their time and effort in handling our paper. We also sincerely thank the four anonymous reviewers for their invaluable comments and suggestions, which have greatly helped us improve the quality of our work.

We have undertaken a major revision of the manuscript and are pleased to resubmit it for your consideration. In this revised version:

- 1) We have addressed all the concerns raised by the reviewers. Specifically, we have revised or rewritten the relevant sections based on their suggestions. Detailed responses to each reviewer's comments are provided below, with corresponding modifications highlighted in blue in the revised manuscript.
- 2) We have incorporated additional references, clarified potential ambiguities, added new experiments, and included a systematic discussion of related works.
- 3) We have carefully polished the language and improved the clarity of figures to ensure accurate and precise presentation.
- 4) We have also included a comprehensive supplementary document, which provides additional implementation and evaluation details beyond the main manuscript.

The detailed responses to the reviewers' comments are attached. We would be happy to incorporate any further feedback or suggestions into the final version of the manuscript.

RESPONSE TO REVIEWER #2

We sincerely thank the reviewer for the invaluable comments. Below are our responses.

Comment 1: *It would be valuable to discuss how the proposed framework could be extended to human-robot collaboration scenarios, where ensuring both robot and human safety is critical.*

[Response:] We have added a discussion in the introduction section on how the proposed framework can be extended to human-robot collaboration scenario. In such scenario, robots not only need to avoid known dangers in the environment, but also be able to understand unseen environments and infer potential consequences. Our SAFETY-AS-POLICY generates complex scenarios that involve both humans and robots in a virtual world model. Through the cognitive learning loop, the LMM gradually accumulates knowledge about human activity risks, demonstrating strong generalization capability. This mechanism provides a foundation and direction for achieving the smart safety in human-robot collaboration.

Comment 2: *The paper is missing a discussion of related work that focuses on safety in robotic manipulation. Please highlight the key differences between your approach and prior methods, especially those that address risk-aware planning or responsible decision-making in robotics.*

[Response:] We have added a discussion of related works in robotic manipulation safety, including theory-based safe planning and control algorithms. These methods mainly focus on static planning or training and struggle to comprehensively address complex and evolving implicit risks. Our SAFETY-AS-POLICY combines cognitive learning with virtual world interactions, enabling automatic discovery and avoidance of diverse novel risks in dynamically generated simulation scenarios. This results in flexible and safe manipulation even in unseen scenarios:

In the field of robotics ... difficult to adapt to complex or unseen risks in real-world environments.

Comment 3: *It would strengthen the paper to systematically define the types of safety risks considered, rather than only providing illustrative examples. A clear taxonomy or categorization would help readers understand the scope of “responsible” behavior.*

[Response:] Our categorization and definitions are based on the ISO 12100:2010 safety of machinery principles [1], which we have simplified into three major categories of risks: electrical, fire & chemical, and human-related risks. Specifically, we have included mechanical injuries and ergonomic hazards under the category of human-related risks.

Comment 4: *Clarify how the TAMP module is implemented: (1) How is task planning performed? (2) How are motion plans generated from tasks? (3) How are kinematic, dynamic, and collision constraints handled? Also, elaborate on how the notion of “responsibility” is operationalized in your system—what*

are its formal or practical components?

[Response for TAMP implementation:] We introduced implementation details of manipulation pipeline in the Section I.B of Supplementary Material:

To evaluate a robot’s capability in manipulating the physical world, we develop a trajectory generation pipeline that integrates LMMs. The pipeline consists of four key modules: perception, task decomposition, generation of affordance and avoidance maps, and motion planning.

In the perception module, YOLO-WORLD [2] and EFFICIENTViT-SAM [3] are employed for open-vocabulary object detection and segmentation. During task planning, LMM interprets human language commands and generates safe action sequences, effectively decomposing natural language instructions into sub-tasks. Leveraging the code generation capabilities of the LMM, 3D affordance and avoidance maps are produced. These maps guide the motion planner in generating efficient and collision-free trajectories. Specifically, the motion planning is based on the VoxPoser framework [4], which leverages the affordance and avoidance maps to inform voxel-based goal proposals and ensure trajectory optimization in complex environments.

To enhance the system’s responsiveness in dynamic environments, the perception and planning modules operate in parallel. When object displacement is detected in the scene, the LMM re-triggers task planning to adapt to the updated environment. Currently, the perception module runs at approximately 8–10 FPS, while the motion planner operates at around 3 FPS. Due to occasional frame-level detection dropouts in the perception module, the system uses the most recently recognized object positions to update subsequent computations and maintain continuity in downstream processing.

[Response for responsibility operationalization:]

In our system, the concept of responsible robotic manipulation is not merely an ethical consideration, but is operationalized as a concrete design objective that is computationally and behaviorally enforced throughout the manipulation pipeline.

We propose SAFETY-AS-POLICY, a framework that transforms the notion of responsibility into actionable modules, ensuring both the safety and reliability of task execution under diverse scenarios. Our work addresses a key challenge: while language instructions provide a flexible and intuitive interface for humans to command robots, they can pose significant safety risks in real-world contexts. Therefore, the robot must not only interpret the instruction but also possess a cognitive understanding of risk associated with the environment and adapt its execution strategy accordingly. To achieve this, we utilize a large multimodal model (LMM) f , which takes as input the visual context v , a human language instruction c , and an explicit representation of risk cognition r . The model then generates task and motion planning (TAMP) code l that is aligned with safety constraints: $l = f(v, c \mid r; p_{\text{imp}})$, where p_{imp} denotes the prompt

template for code generation. These TAMP codes employ predefined low-level action APIs such as move, rotate, or tilt, enabling the robot to perform the task while adhering to safety-aware execution plans driven by the understanding of environmental risks.

Overall, responsibility is implemented through the following system-level mechanisms:

- 1) Safety-aware TAMP pipeline using LMM.
- 2) The integration of affordance and avoidance maps to define safe operational zones.
- 3) Voxel-level motion planner to generate trajectory from affordance and avoidance maps generated from LMM.
- 4) Replanning in response to dynamic scene changes.

In summary, responsibility in our framework is not an abstract ideal, but a principled, enforceable design paradigm embedded into the perception, reasoning, and execution layers of robotic manipulation.

Comment 5: *Please compare your approach to prior work that explicitly considers safety or risk in robotic manipulation to better contextualize the effectiveness of your method.*

[Response:] Our FILTER-AND-RETRY strategy was inspired by filter-based methods which explicitly consider the risk in robotics [5]. We have added appropriate citations and discussion. Unlike prior works, our method not only covers traditional explicit constraints but also adapts to implicit and novel risks via reasoning. Experimental results demonstrate that both safety rate and task completion rate are significantly superior to existing approaches.

Comment 6: *Including video demonstrations in the supplementary material would be highly beneficial for evaluating both functionality and safety aspects of the system.*

[Response:] We have included links to video demonstrations in the Section XIII of Supplementary Material to facilitate evaluation of both system functionality and safety aspects. For more details, please visit our homepage: <https://sites.google.com/view/safety-as-policy-anonymous>.

RESPONSE TO REVIEWER #5

We appreciate the reviewer’s recognition of our work. We will continue to improve our paper!

RESPONSE TO REVIEWER #12

We sincerely thank the reviewer for the detailed comments. Below are our responses.

Comment 1: *In related work part, the paper lacks a clear explanation of when and why traditional methods fail. For example, pre-task checks and visual safety assessments are mentioned, but their limitations in real-time manipulation are not discussed. Please clarify their shortcomings, especially*

in terms of handling unobservable or evolving risks during execution.

[Response:] Traditional pre-task checks and visual safety assessments are limited in handling unobservable or dynamically evolving risks during execution. Our cognitive reasoning model actively anticipates and avoids such future hazards through scene reasoning and virtual interaction, enhancing both generalization and practicality. We have added this discussion in the related work section:

In the field of robotics ... difficult to adapt to complex or unseen risks in real-world environments.

Comment 2: *Dangers in the real world can be highly diverse (we can image the water in the bottle can be various kinds of liquids, or it can be many different things in a closed box). The paper does not explain how the model remains stable and effective across such variability. Consider adding discussion on this.*

[Response:] The robustness comes from the LMM’s rich world knowledge and strong generalization capabilities. For example, knowledge about “pesticide bottles” can generalize to various toxic substances without retraining. This enables our method to remain effective in unseen scenarios such as unknown liquids or sealed boxes.

We validate our pipeline with different components across different tasks to check the performance for task scene with different variants as shown in Fig. B of Supplementary Materials, e.g., using watering can or cup for flower watering task. For more details, please check out more experiments results in Supplementary Materials.

Comment 3: *The logic of Fig. 4 is unclear. Relationships between components are hard to follow. Please consider reorganizing the figure and providing clearer descriptions of each step. Figs. 4 and 5 should align more explicitly with the equations in the text. Label figure components using symbols or equation numbers where appropriate.*

[Response:] We have revised the figures related to the world model and cognition learning pipeline with detailed labeling of symbols and explicit correspondence to the equations in the text. To avoid confusion, please note that due to a numbering error in the original submission, the figures have been renumbered in the revision: original Fig. 4 → new Fig. 3, original Fig. 5 → new Fig. 4.

Comment 4: *As a core contribution, the SafeBox dataset needs more detail. Please include its scenario types, parameter design, intended use cases, and any constraints.*

[Response:] Detailed descriptions of SafeBox scenario types, parameter design, intended use cases, and constraints have been included in Sections II and III of Supplementary Materials.

Comment 5: *The Filter and Retry strategy lacks citations. If based on prior work, please include references. If novel, explain how it differs from similar methods.*

[Response:] Inspired by the filter-based methods which explicitly consider the risk in robotics [5], we also design FILTER-AND-RETRY as a reference baseline. We have added relevant citations in the revised version.

Comment 6: *Too many simulated danger cases may lead to over-conservative behavior. Please discuss how the model avoids overestimating risk and ensures practical task completion.*

[Response:] LMM-based task planning already demonstrates strong task completion ability but lacks safety awareness [6]. Our SAFETY-AS-POLICY addresses this gap. As shown in Section VI of Supplementary Material, the cognitive planning module does not lead to over-conservatism. Instead, as the number of cognitive samples increases, task success rates steadily improve because reasoning can generalize across similar cases, avoiding over-conservative behavior.

Comment 7: *To better highlight real-time risk detection, consider adding an experiment where a dangerous object appears during execution. This would contrast with traditional pre-check methods and strengthen your claims.*

[Response:] Thank you for the suggestion. While our current work focuses on risks emerging during execution, we recognize the importance of real-time risk detection and response. Therefore, we modified our pipeline with a simple LMM-based discriminator that detects environmental changes requiring re-planning. If triggered, the system pauses and replans actions. Experiments in the Section X of Supplementary Material demonstrate that our method can effectively handle such scenarios. For more details, please visit our homepage: <https://sites.google.com/view/safety-as-policy-anonymous>.

RESPONSE TO REVIEWER #13

We sincerely thank the reviewer for the constructive comments. Below are our responses.

Comment 1: *Only 10 real-world tasks were tested, raising questions about generalizability. More complex environments (e.g., dynamic crowds) are untested.*

[Response:] All of our real-world tests were unseen scenarios, demonstrating generalization to new tasks and high-risk conditions. Previous LMM-based works [6, 7, 8] also show transferability and scalability. Additionally, we adapted our pipeline with an LMM-based discriminator for dynamic crowds, which pauses and replans actions upon detecting environmental changes. Experiments in the Section X of Supplementary Material confirm effectiveness and generalization in such complex scenarios.

Comment 2: *SafeBox relies on DALL-E-3 renders, which may not capture real-world noise (e.g., lighting, occlusion).*

[Response:] As described in Section I of the Supplementary Materials, the perception module of our model in the real world uses YOLO-WORLD and EFFICIENTViT-SAM without any fine-tuning. DALL-E-3 is only used for acquiring text-level cognition of risk scenarios to generate TAMP code, rather than for learning general manipulation. This design helps avoid the sim-to-real gap. Furthermore, DALL-E-3 can synthesize diverse environments, *e.g.*, different lighting or occlusions, and our experiments in real-world environments also confirm the effectiveness of the cognition learned from the virtual world.

Comment 3: *GPT-4o + DALL-E-3 usage could be prohibitive for low-resource settings.*

[Response:] We only need GPT-4o and DALL-E-3’s API, which impose no additional computational burden on robots. Moreover, open-sourced LMMs such as Qwen-VL can replace such models, ensuring scalability and applicability in low-resource settings [9, 10, 11].

Comment 4: *Risks are limited to electrical/fire/chemical/human. The other hazards, e.g., mechanical, privacy, are ignored. That is, the task categorization lacks broader safety domains.*

[Response:] Our categorization and definitions are based on the ISO 12100:2010 safety of machinery principles [1], which we have simplified into three major categories of risks: electrical, fire & chemical, and human-related risks. Specifically, we have included mechanical injuries and ergonomic hazards under the category of human-related risks.

Comment 5: *The mental model’s learnable prompt lacks interpretability, hindering the analysis of how cognition evolves. .*

[Response:] We provide the full learned cognition prompt in Section VIII of Supplementary Materials, enabling clear traceability of cognitive evolution and final safety knowledge. This supports interpretability and analysis of the model’s cognitive learning process.

- [1] International Organization for Standardization, Safety of machinery – General principles for design – Risk assessment and risk reduction, Std. ISO 12100:2010, 2010, includes amendment ISO 12100:2010/Amd.1:2011.
- [2] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, “Yolo-world: Real-time open-vocabulary object detection,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 16 901–16 911.
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- [4] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, “Voxposer: Composable 3d value maps for robotic manipulation with language models,” in 7th Annual Conference on Robot Learning, 2023.
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- [7] P. Liu, Y. Orru, J. Vakil, C. Paxton, N. M. M. Shafiullah, and L. Pinto, “Ok-robot: What really matters in integrating open-knowledge models for robotics,” arXiv preprint arXiv:2401.12202, 2024.
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- [11] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang et al., “Qwen2. 5-vl technical report,” arXiv preprint arXiv:2502.13923, 2025.